

# [Lecture Notebook] The Purpose of Research

GE2021 Research Technology, Spring 2025

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## Learning Objectives

|   | X                 | mpg  | cyl | disp | hp  | drat | wt    | qsec  | vs | am | gear | carb |
|---|-------------------|------|-----|------|-----|------|-------|-------|----|----|------|------|
| 1 | Mazda RX4         | 21.0 | 6   | 160  | 110 | 3.90 | 2.620 | 16.46 | 0  | 1  | 4    | 4    |
| 2 | Mazda RX4 Wag     | 21.0 | 6   | 160  | 110 | 3.90 | 2.875 | 17.02 | 0  | 1  | 4    | 4    |
| 3 | Datsun 710        | 22.8 | 4   | 108  | 93  | 3.85 | 2.320 | 18.61 | 1  | 1  | 4    | 1    |
| 4 | Hornet 4 Drive    | 21.4 | 6   | 258  | 110 | 3.08 | 3.215 | 19.44 | 1  | 0  | 3    | 1    |
| 5 | Hornet Sportabout | 18.7 | 8   | 360  | 175 | 3.15 | 3.440 | 17.02 | 0  | 0  | 3    | 2    |
| 6 | Valiant           | 18.1 | 6   | 225  | 105 | 2.76 | 3.460 | 20.22 | 1  | 0  | 3    | 1    |

| X                | mpg           | cyl           | disp          |
|------------------|---------------|---------------|---------------|
| Length:32        | Min. :10.40   | Min. :4.000   | Min. : 71.1   |
| Class :character | 1st Qu.:15.43 | 1st Qu.:4.000 | 1st Qu.:120.8 |
| Mode :character  | Median :19.20 | Median :6.000 | Median :196.3 |
|                  | Mean :20.09   | Mean :6.188   | Mean :230.7   |
|                  | 3rd Qu.:22.80 | 3rd Qu.:8.000 | 3rd Qu.:326.0 |
|                  | Max. :33.90   | Max. :8.000   | Max. :472.0   |

  

| hp            | drat          | wt            | qsec          |
|---------------|---------------|---------------|---------------|
| Min. : 52.0   | Min. :2.760   | Min. :1.513   | Min. :14.50   |
| 1st Qu.: 96.5 | 1st Qu.:3.080 | 1st Qu.:2.581 | 1st Qu.:16.89 |
| Median :123.0 | Median :3.695 | Median :3.325 | Median :17.71 |
| Mean :146.7   | Mean :3.597   | Mean :3.217   | Mean :17.85   |
| 3rd Qu.:180.0 | 3rd Qu.:3.920 | 3rd Qu.:3.610 | 3rd Qu.:18.90 |
| Max. :335.0   | Max. :4.930   | Max. :5.424   | Max. :22.90   |

  

| vs             | am             | gear          | carb          |
|----------------|----------------|---------------|---------------|
| Min. :0.0000   | Min. :0.0000   | Min. :3.000   | Min. :1.000   |
| 1st Qu.:0.0000 | 1st Qu.:0.0000 | 1st Qu.:3.000 | 1st Qu.:2.000 |
| Median :0.0000 | Median :0.0000 | Median :4.000 | Median :2.000 |
| Mean :0.4375   | Mean :0.4062   | Mean :3.688   | Mean :2.812   |
| 3rd Qu.:1.0000 | 3rd Qu.:1.0000 | 3rd Qu.:4.000 | 3rd Qu.:4.000 |
| Max. :1.0000   | Max. :1.0000   | Max. :5.000   | Max. :8.000   |

34 **Reference**